

HOLMES CTF 2026

GLOBAL THREAT INTELLIGENCE
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FORENSICS
THREAT INTELLIGENCE
DIGITAL INVESTIGATION
TRUTH PERSISTS

LONDON

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THE REICHENBACH DIRECTIVE

A Complete DFIR Investigation
of Operation DIOGENES

NODE-LINK ANALYSIS
ARTIFACT CORRELATION
TIMELINE RECONSTRUCTION
IOC HUNTING

MichelDV · InFracturaVeritas
Exploit Bag Chaser

104 / 111 Flags · 8,200 Points
Rank 211 / 5,637 Teams

0000	47 45 54 20 2f 61 70 69	GET /api
0010	48 54 54 50 2f 31 2e 31	HTTP/1.1
0020	48 6f 73 74 3a 20 64 69	Host: di
0030	6f 67 65 6e 65 73 2e 69	ogenes.i
0040	0d 0a 55 73 66 72 2d 41	.User-AA

TCP 443

TCP 80

TCP 22

DNS

TLS 1.3

PACKET FLOW ANALYSIS
ANOMALY DETECTED

RECONNAISSANCE

INITIAL ACCESS

EXECUTION

PERSISTENCE

PRIVILEGE ESCALATION

DEFENSE EVASION

CREDENTIAL ACCESS

LATERAL MOVEMENT

OBJECTIVE

TARGET: LONDON
OPERATION DIOGENES

NINE INVESTIGATIONS. ONE OPERATION.

Operation DIOGENES reconstructed across endpoint, network, blockchain, mobile, Active Directory, malware and flight telemetry evidence.

EVIDENCE → CORRELATION → RECONSTRUCTION



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This publication consolidates nine separate HOLMES CTF 2026 Sherlock investigations into a single cross-case forensic case study. Exact evidentiary values are retained where they materially support a finding; repetitive local workflow details are minimized.

Executive Summary

The Reichenbach Directive presented nine investigations that initially appeared to occupy different technical domains: Windows endpoint forensics, XMPP infrastructure, removable-media compromise, Linux supply-chain intrusion, Android malware, UAV telemetry, Active Directory abuse, memory forensics and blockchain-assisted theft.

Taken together, the evidence describes a broader operation centered on **DIOGENES** and repeatedly associated with **NAPOLEON** infrastructure, operator identities, staging nodes, physical access, credential abuse, covert communications and post-compromise cleanup.

The nine cases can be understood as several connected operational threads rather than nine isolated puzzles.

The first thread begins with **Silent Dividend**, where a malicious Electron application combined a LuaJIT implant with Ethereum Sepolia smart contracts. The malware monitored wallet-related secrets and used on-chain state to obtain a decryption key. The contract chain also exposed the geographic pivot **51.5049, 0.0348**, placing the operation in **Silvertown**.

A second thread centers on **Watson's abduction and the Murknet XMPP environment**. **Whisper Chain** recovered member credentials from OPSEC failures, exposed hidden operational rooms, decrypted all 18 pubsub command items and reconstructed **Operation Snatch**. The custody affiliate **spur** connects directly to **Bottle Out**, where a seized laptop revealed Tactical RMM, OpenVPN, Gajim credentials, Operation Vanish and the operator identity **Abel Stokes**.

A third thread shows DIOGENES personnel being targeted through physical and software supply-chain compromise. **Paper Ghost** reconstructed a planted USB attack against **C0-LT-0469**, recovered microphone and webcam surveillance activity, and extracted sensitive DIOGENES records from **Windows.edb** using custom ESENT parsing. Those records exposed developer **Tom Ainsworth** and his ticketing credentials. **Poisoned Branch** then showed Ainsworth compromised through a poisoned **diogenes-ticket-parser** repository, a runtime-reconstructed Mettle implant, operator infrastructure at **BlackPearl2026.htb**, and an encrypted **L00T.zip** recovered without its password through a ZipCrypto known-plaintext attack.

A fourth thread moves from ground infrastructure to mobile and aerial platforms. **Silent Passenger** reconstructed a staged Android compromise inside an Allwinner T3/TOPWAY vehicle head unit. A privileged TWCore application used MQTT to deliver **com.tw.jar1**, which reconstructed additional stages, retrieved RSA-protected tasking and ultimately loaded **sh66.io**, turning the vehicle into a **mobile proxy relay**. **Iron Feather** then recovered the mission and real flight of a PX4 quadcopter launched from the same **Pavilion Road** area. Its dataman and ULog evidence had been protected with a custom AES-256-GCM format and KDF. After decryption, the mission showed a payload release near Hyde Park / Rotten Row and a return flight interrupted by an injected **MAV_CMD_INJECT_FAILURE**, ending with an impact on Knightsbridge.

The final thread is the enterprise compromise. **Borrowed Name** reconstructed an intrusion from removable media on **WS02**, DLL side-loading, AdaptixC2, Active Directory discovery, Internal Monologue-style credential access, ResetNightmare abuse and lateral movement to **DC02**. The attacker uploaded **svc_bkup** through **ADMIN\$** and executed it through a temporary service-path modification. **Last Light** begins with that same binary already resident on the Domain Controller. The exact **svc_bkup** artifact was reconstructed from memory and matched the earlier file. From there, memory and event-log evidence exposed token impersonation, a forged root-domain TGT, machine-account creation, Resource-Based Constrained Delegation and an S4U-derived **cifs/DC02** ticket.

The resulting portfolio demonstrates not one narrow toolset, but the ability to move between evidence types while maintaining a consistent evidentiary standard:

- filesystem and deleted-file recovery;
- Windows Registry, SRUM, ConsentStore and Search Index analysis;
- XMPP/MAM/pubsub investigation;
- malware and protocol reverse engineering;
- smart-contract and wallet-drainer analysis;
- Linux endpoint and operator-server forensics;
- Android firmware and staged payload reconstruction;
- custom cryptographic KDF recovery;
- PX4 dataman and ULog reconstruction;
- packet analysis and C2 reverse engineering;
- Active Directory and Kerberos abuse analysis;
- memory forensics and cross-case artifact validation.

The final result was **104 validated flags out of 111**, with **six of nine Sherlocks fully solved**.

1. Investigation Scope and Methodology

1.1 Evidence handling

Original evidence was treated as immutable. Analysis was performed on working copies in a Kali Linux virtual machine using a dedicated DFIR, reverse-engineering, network-analysis and malware-analysis toolchain.

The governing principle throughout the investigation was:

Reproducible enough to trust the findings, not exhaustive enough to reproduce the analyst's entire workflow.

This distinction matters. A forensic report should expose the chain of evidence necessary to understand and validate a conclusion, but it does not need to preserve every exploratory command, failed hypothesis or local implementation detail.

1.2 Analytical model

Findings in this report fall into four categories:

Label	Meaning
FACT	Directly supported by one or more artifacts
CORRELATION	Supported by multiple artifacts or cases whose relationship is independently observable
INFERENCE	A reasoned interpretation consistent with the evidence but not directly encoded in a single artifact
OPEN	Technically reconstructed evidence exists, but a challenge value remains unvalidated or incomplete

Platform acceptance is treated as an additional validation signal, not as a substitute for forensic reasoning.

1.3 Time normalization

Where timestamps were available, they were normalized to UTC in the underlying case reports. Some investigations, especially Silent Passenger and Iron Feather, contain relative or simulator-derived time rather than a complete campaign calendar timestamp. Those events are preserved as relative sequences instead of being forced into a false absolute chronology.

1.4 Cross-validation

The strongest findings in the investigation are those that survive independent validation.

Examples include:

- `.integrity` on Tom Ainsworth's host matching `reverse.bin` on the operator server;
- `svc_bkup` recovered from DC02 memory matching the binary transferred in Borrowed Name;
- the Iron Feather AES key validating the GCM authentication tag on both encrypted evidence containers;
- Paper Ghost's USB path appearing across UserAssist, ConsentStore and SRUM;
- Whisper Chain's member credentials emerging from two independent OPSEC leaks rather than brute force;
- Bottle Out's deleted VPN/XMPP data being reconstructed from surviving NTFS and application artifacts after Operation Vanish.

1.5 Investigation outcome

Sherlock	Difficulty	Validated	Primary discipline
S01 — Silent Dividend	Medium	10/10	Windows / Electron / blockchain
S02 — Bottle Out	Easy	10/10	Windows endpoint / XMPP / anti-forensics

Sherlock	Difficulty	Validated	Primary discipline
S03 — Whisper Chain	Medium	7/8	XMPP / threat intelligence
S04 — Paper Ghost	Hard	9/9	Windows DFIR / search-index recovery
S05 — Poisoned Branch	Hard	15/15	Linux / supply chain / loot recovery
S06 — Silent Passenger	Hard	16/20	Android / staged malware / proxying
S07 — Iron Feather	Hard	17/17	UAV / cryptography / flight forensics
S08 — Borrowed Name	Insane	12/12	Active Directory / C2 / lateral movement
S09 — Last Light	Insane	8/10	Memory / Kerberos / RBCD
Total		104/111	

Seven tasks remain open: Whisper Chain T8; Silent Passenger T13, T14, T19 and T20; Last Light T2 and T3.

2. The Reichenbach Directive — Incident Overview

The evidence is best understood as a set of operational threads converging on DIOGENES personnel, infrastructure and access.

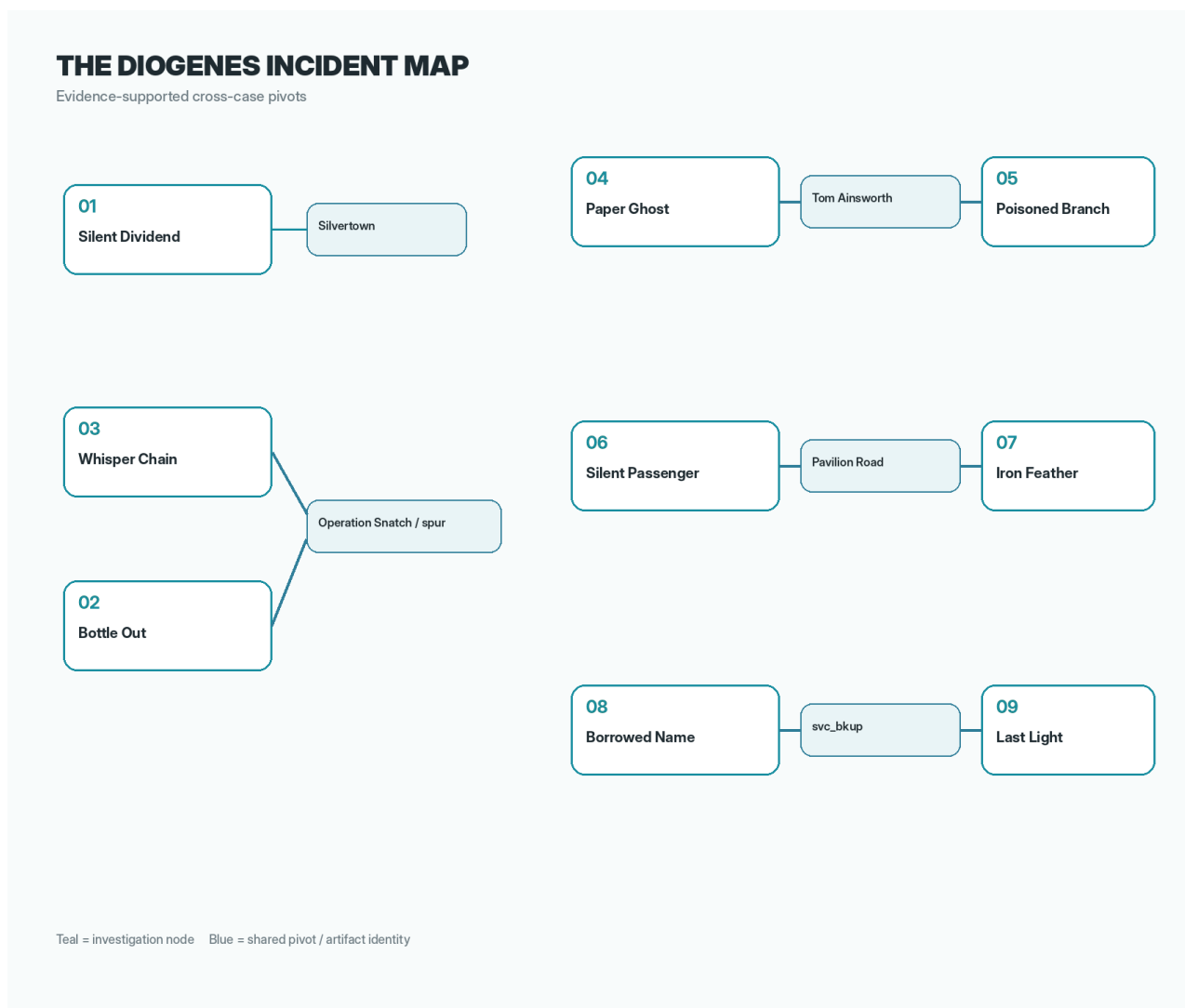


Figure — The DIOGENES incident map

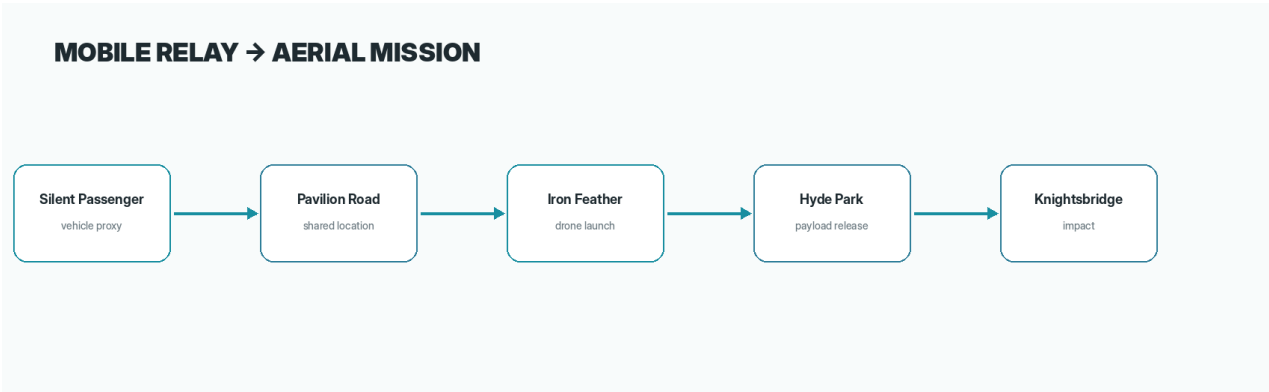


Figure — Mobile relay to aerial mission chain

2.1 Strongest cross-case links

Cases	Correlation	Confidence
S03 → S02	<code>spur</code> is the custody affiliate in Operation Snatch and the operator context on the seized laptop; Murknet appears in both cases	High
S04 → S05	Paper Ghost exposes Tom Ainsworth and DIOGENES ticketing credentials; Poisoned Branch compromises Ainsworth through a ticket-parser supply-chain lure	High
S06 → S07	Pavilion Road is the location associated with the compromised vehicle and the drone's launch/home position	High
S08 → S09	<code>svc_bkup</code> transferred to DC02 in Borrowed Name is reconstructed from memory in Last Light with matching artifact identity	Very High
S01 → wider campaign	<code>AUTH=NAPOLEON</code> , settlement reference and Silvertown coordinates align with the broader operation	High

The report deliberately avoids forcing weaker narrative relationships where artifact-level support is absent.

3. Master Incident Timeline

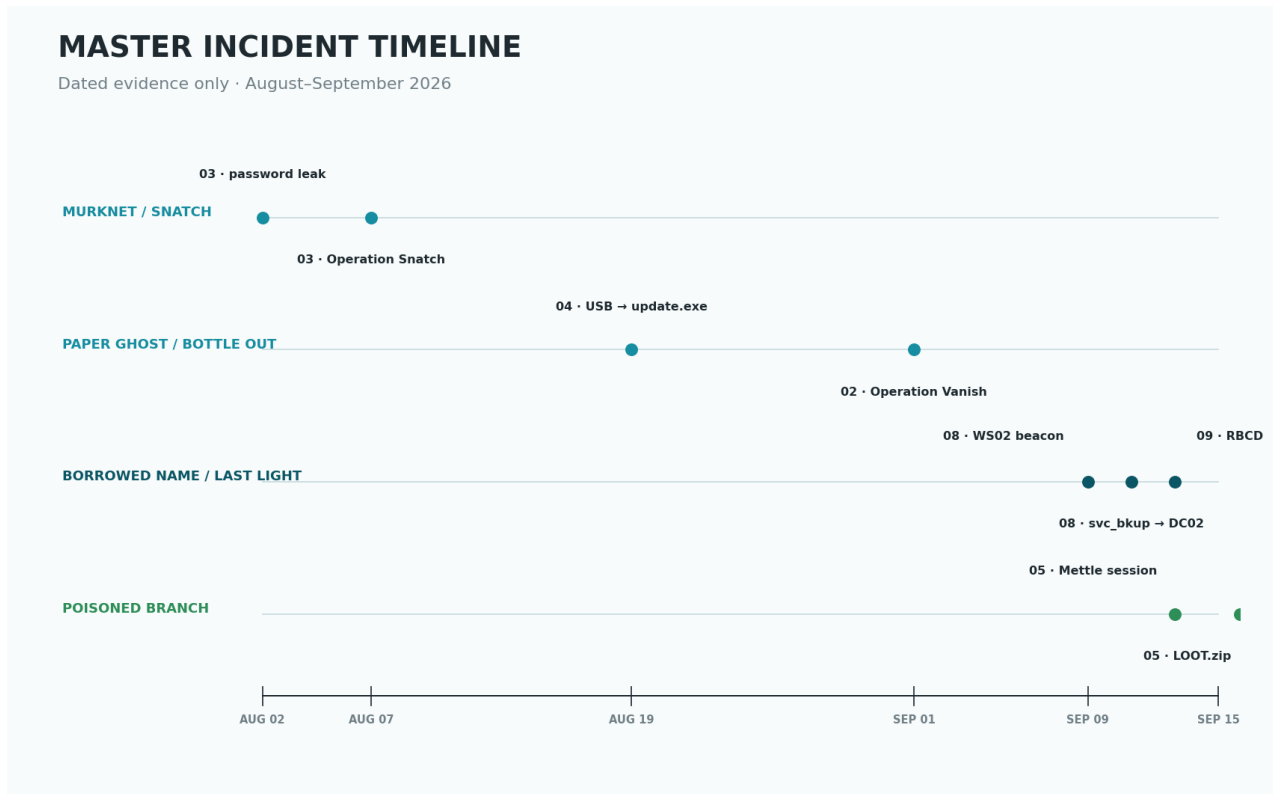


Figure — Dated campaign evidence from August–September 2026

The cases do not all provide a single absolute calendar chronology. The timeline below therefore separates dated evidence from relative operational chains.

3.1 Dated evidence

Date / Time (UTC)	Case	Event
2026-08-02 16:30	Whisper Chain	Historical temporary passwords exposed in Murknet Infrastructure
2026-08-02 18:47	Whisper Chain	Onboarding PDF published with zytglogge88@murknet.htb in metadata
2026-08-04 13:20	Whisper Chain	Operation Sparkling begins
2026-08-06 23:22	Whisper Chain	Operation Snatch dispatcher key rotates to SHALLOWBLUE
2026-08-07 07:38	Whisper Chain	Operation Snatch begins
2026-08-07 10:50	Whisper Chain	dynamite starts the operation against Watson
~2026-08-07 12:33	Whisper Chain	Watson delivered to spur at container 75JM77 in Silvertown
2026-08-07 20:18	Whisper Chain	spur reports the site discovered and flees
2026-08-07 20:21	Whisper Chain	Immediate cleanup ordered
2026-08-19 15:35:50	Paper Ghost	Lexar USB registered on C0-LT-0469
2026-08-19 15:36:25	Paper Ghost	update.exe executed from removable media
2026-08-19 15:38:08	Paper Ghost	Microphone capture begins
2026-08-19 15:42:28	Paper Ghost	Webcam capture begins

Date / Time (UTC)	Case	Event
2026-09-01 08:33	Bottle Out	Windows Security log cleared
2026-09-01 12:37:40–12:37:47	Bottle Out	Operation Vanish removes Gajim/OpenVPN/VPN artifacts
2026-09-09 20:32:33	Borrowed Name	AdaptixC2 beacon checks in from WS02
2026-09-09 20:44:11	Borrowed Name	Internal Monologue-style NetNTLMv2 extraction
2026-09-09 20:46:47	Borrowed Name	ResetNightmare resets jreed password
2026-09-09 20:49:36	Borrowed Name	svc_bkup uploaded to DC02 and defragsvc modified
2026-09-09 20:49:57	Borrowed Name	SYSTEM beacon established from DC02
2026-09-09 22:19:51	Last Light	Forged root-domain TGT observable in LSASS
2026-09-09 22:35:44	Last Light	Machine account svc_bkup\$ created
2026-09-09 22:38:41	Last Light	RBCD attribute written on DC02
2026-09-09 22:39:43	Last Light	S4U-derived cifs/DC02 service ticket observed
2026-09-09 22:40:38	Last Light	DC02 memory acquired
2026-09-15 15:20:21	Poisoned Branch	Mettle handler/session active against Tom Ainsworth
~2026-09-15 15:40	Poisoned Branch	HR roster collected
2026-09-15 15:40:57	Poisoned Branch	HR roster deleted from victim
2026-09-15 15:43	Poisoned Branch	L00T.zip created on operator infrastructure

3.2 Relative operational chain

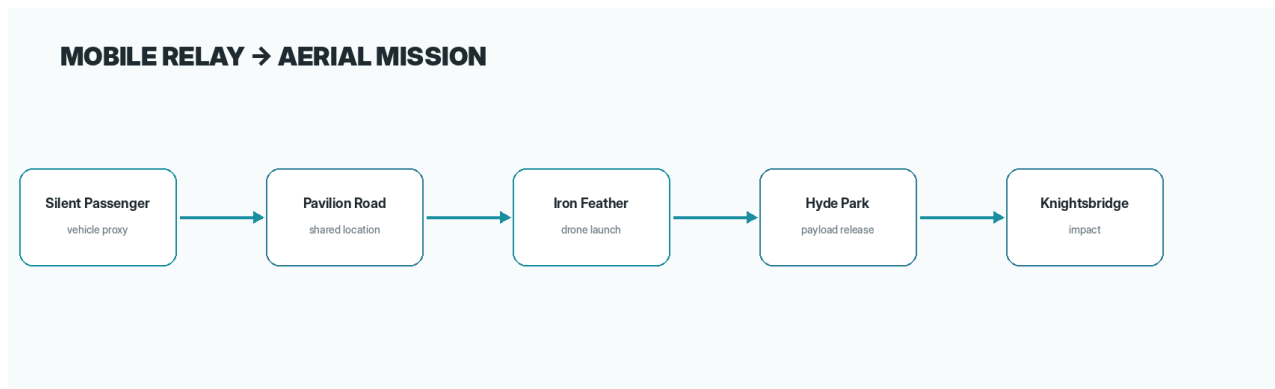


Figure — Mobile relay to aerial mission chain

Silent Dividend also contributes a geographic pivot into Silvertown but does not provide a campaign timestamp suitable for placement into the dated sequence.

CASE 01

01

SILENT DIVIDEND

Electron Malware · On-Chain Key Retrieval · Wallet
Draining

ELECTRON / LUAJIT / SEPOLIA / WINHTTP

10 / 10

VALIDATED



CASE 01

SILENT DIVIDEND

Electron Malware, On-Chain Key Retrieval and Wallet Draining

Case Context

Silent Dividend opens the investigation with a hybrid endpoint-and-blockchain attack. The apparent application, `TrustSettle 1.0.0.exe`, is an Electron settlement client. Its user-facing behavior is plausible, but its preload script also has direct access to Node.js capabilities because the application runs with permissive Electron settings.

Evidence

The preload stages `api.txt`, `luajit.exe`, `lua51.dll` and `.env` into `C:\Users\Public`, then launches the Lua implant through hidden PowerShell.

The LuaJIT implant uses FFI rather than a conventional native wrapper. It loads `kernel32` and `winhttp`, monitors filesystem changes through `ReadDirectoryChangesW`, interprets the resulting buffer as `FILE_NOTIFY_INFORMATION`, and opens `C:\Users\Public\.env` for reading.

The `.env` lure is particularly significant because the supplied instructions direct the user to populate it with wallet-related values such as:

```
PRIVATE_KEY
WALLET_ADDRESS
RPC_URL
```

Remote transmission is implemented through the WinHTTP stack, including `WinHttpSendRequest`.

On-chain control path

The Electron preload also queries the Sepolia `StateRegistry` contract:

```
0xbB63Ae28E4f75C9392bae69cDf5394Ca0ACdA6B1
```

through:

```
resolveState()
```

The returned `bytes32` value is the decryption key for an embedded command. Decrypting the payload reveals:

```
AUTH=NAPOLEON SETTLEMENT_REFERENCE=SR-4821
```

and opens `%TEMP%\settlement.html`.

The settlement page uses `ethers.BrowserProvider`, requests an unlimited ERC-20 allowance through `approve()`, and sets the amount to `MaxUint256`.

The associated drainer contract:

```
0x69Bf5b7aBA51C3Ee8bF169aB47479ba95DBF709D
```

derives its hidden owner as:

```
0xEBfC1eD96b1C6b940fb6B06359fF4A6776Df7a9A
```

The same contract path exposes:

```
51.5049,0.0348
```

which places the operation in Silvertown.

Key Finding

Silent Dividend is not simply wallet phishing. The endpoint implant, on-chain key retrieval and approval-drainer flow are mutually supporting components of a single attack chain.

Detection Opportunities

- alert on hidden PowerShell spawning `luajit.exe`;
- monitor execution or script content from `C:\Users\Public`;
- detect non-browser processes reading wallet `.env` files;
- monitor unexpected WinHTTP activity from LuaJIT;
- enforce Electron hardening such as `contextIsolation`;
- inspect high-risk ERC-20 allowance requests, especially `MaxUint256`;
- monitor enterprise endpoints interacting with unapproved blockchain RPC services where relevant.

Validation

10/10 tasks validated.

CASE 02

02

BOTTLE OUT

Windows Endpoint · XMPP Artifacts · Anti-Forensics

E01 / NTFS / OPENVPN / XMPP

10 / 10

VALIDATED



CASE 02

BOTTLE OUT

Windows Endpoint and XMPP Artifact Forensics

Case Context

Bottle Out examines the laptop associated with Watson's jailer. The host was configured as an operational endpoint rather than a conventional workstation.

Evidence

Three communication layers were visible:

- **Tactical RMM Agent v.2.11.0** connected to `api.antimattercommunication.xyz`;
- **OpenVPN** connected to `18.156.81.166:7577`, with CA `NPLN-CA` and assigned address `10.129.175.2`;
- **Gajim Portable 2.6.0** used the account `spurio9@murknet.htb` with password `spur999!*`.

The laptop therefore connects directly to the Murknet infrastructure explored in Whisper Chain.

Browser Local Storage exposed the operator identity associated with the laptop:

```
Abel Stokes  
Abel.stokes@hotmail.com
```

Operation Vanish

At 12:37 UTC on 1 September 2026, an encoded PowerShell cleanup routine recursively deleted Gajim, OpenVPN and VPN directories.

The first accepted command was:

```
Remove-Item -LiteralPath C:\Users\spur\Gajim -Recurse -Force
```

Deletion did not destroy the investigation. Deleted NTFS records, surviving dataruns, registry state, PowerShell logging, application databases and browser storage were sufficient to recover the core communications stack and identity.

A separate Security log-clearing event earlier in the day further reduced visibility but did not remove the principal evidence.

Cross-case Significance

Bottle Out provides the physical endpoint behind the `spur` identity seen in Whisper Chain. It is the strongest endpoint-side anchor for the Watson custody thread.

Detection Opportunities

- alert on newly installed remote-management agents on short-lived or operational workstations;
- correlate OpenVPN deployment with newly created local accounts;
- monitor Gajim or other XMPP clients in environments where they are not authorized;
- detect recursive PowerShell deletion of communication/VPN directories;
- alert on Security log clearing and correlate it with subsequent administrative activity;
- preserve NTFS metadata and USN Journal evidence during containment.

Validation

10/10 tasks validated.

CASE 03

03

WHISPER CHAIN

XMPP Threat Intelligence · Encrypted Command Infrastructure

XMPP / MAM / PUBSUB / AES-CBC



7 / 8

VALIDATED

CASE 03

WHISPER CHAIN

XMPP Threat Intelligence and APT Coordination

Case Context

Whisper Chain moves from endpoint artifacts into the APT's live coordination environment. The meaningful evidence was hosted on [murknet.htb](#) and its related XMPP components.

Infrastructure

The self-signed certificate identified:

```
murknet.htb
groups.murknet.htb
command.murknet.htb
upload.murknet.htb
```

Public rooms were:

```
Infrastructure
Random
Resources
Rules
```

The public message archive contained 466 unique messages.

Credential recovery through OPSEC failure

The pivotal access path required no protocol exploit and no brute force.

A historical password list leaked:

```
KillBill2025!
K4w4Bong424!
Northwind225!
TickTock24!
```

Meanwhile, metadata in [Operational_Onboarding_Guide_v3.2.pdf](#) exposed:

```
Author   = zytglogge88@murknet.htb
Creator  = swissclock
```

Correlating the [swissclock](#) identity with the leaked password [TickTock24!](#) produced valid credentials:

```
zytglogge88@murknet.htb:TickTock24!
```

Hidden operations

Authenticated access exposed:

```
Operation Sparkling
Operation Snatch
```

Operation Snatch coordinated Watson's surveillance, abduction, transportation and detention.

[dynamite](#) handled transport, while [spur](#) took custody. The evidence placed Watson in container:

```
75JM77
```

with watchword:

Chaos Is Order

The command dispatcher used encrypted pubsub items encoded as Base64 `Salted__` blobs. The encryption path was reconstructed as PBKDF2-HMAC-SHA256 followed by AES-256-CBC. All **18/18 command items** were decrypted.

OSINT correlation

The actor alias `rattlesnake` correlated with the social identity **Porlock**, and the investigation recovered cryptocurrency addresses and infrastructure associated with the wider operation.

Cross-case Significance

`spur` is the custody affiliate in Operation Snatch and the operational identity connected to the laptop in Bottle Out.

The cleanup instruction observed in the command stream is consistent with the later anti-forensic behavior on that endpoint, but the report treats the later Operation Vanish event as corroborating context rather than claiming a direct one-to-one execution of the earlier command.

Detection Opportunities

- inventory and monitor XMPP clients and outbound XMPP/STARTTLS in enterprise environments;
- detect unauthorized access to `groups.*`, `command.*` and `upload.*`-style service components;
- inspect exposed document metadata before publication;
- enforce secret rotation and prevent operational passwords from appearing in chat;
- use infrastructure and account correlation across messaging, endpoint and network telemetry.

Validation

7/8 tasks validated.

Open: **T8** — evidence supports **Operation Snatch / kidnapping Watson**, but the expected validator representation was not identified.

CASE 04

04

PAPER GHOST

Planted USB · Endpoint Surveillance · Search-Index
Recovery

USB / SRUM / CONSENTSTORE / WINDOWS.EDB

9 / 9

VALIDATED



CASE 04

PAPER GHOST

Planted USB, Endpoint Surveillance and Search-Index Recovery

Case Context

Paper Ghost reconstructs a targeted physical intrusion against **CO-LT-0469**, used by **cvoss** in the Ministerial Briefing Office.

The planted device was a Lexar USB registered as:

```
CO-USB-0091
Serial: RS200000000627E4&0
```

The false contractor identity was:

```
Elias Venn
EXT-0431
```

Initial access and execution

The USB presented a plausible workstation-specific support package:

```
E:\CO-LT-0469 update package\update.exe
```

The device registered at approximately 15:35:50 UTC on 19 August 2026. UserAssist records one execution of **update.exe** at 15:36:25.

Surveillance and exfiltration

ConsentStore recorded:

```
Microphone start: 15:38:08
Webcam start:      15:42:28
Webcam duration:   126.9805499 s
```

SRUM attributed:

```
172,064,531 bytes sent
172.064531 MB outbound
```

to the malicious executable.

The supplied triage did not contain packet captures or firewall telemetry, so the exact C2 destination is not asserted.

Windows Search as residual evidence

The case's most important forensic technique was the recovery of DIOGENES content from **Windows.edb**.

The original files were absent, but **System_Search_AutoSummary** retained indexed content as ESENT long values. Standard **Libesedb** handling could not resolve the relevant 8-byte long-value keys.

A custom parser reconstructed the long-value B-tree records and decoded compressed chunks using XPRESS LZ77 and bit-packed data recovery.

Six DIOGENES documents were recovered, including the personnel/access record for:

```
Tom Ainsworth
Software Developer
DIOGENES Ticketing Support
```

and:

```
tainsworth:D10g3n3s_T1ck3ts#2026
```

Cross-case Significance

Paper Ghost creates the personnel and credential bridge into Poisoned Branch. It establishes Tom Ainsworth as a DIOGENES ticketing developer and exposes the access context later targeted through a poisoned ticket-parser repository.

Detection Opportunities

- restrict execution from removable media;
- alert on first-seen USB storage followed immediately by executable launch;
- monitor microphone/webcam ConsentStore activity from unusual non-packaged executables;
- detect anomalous high-volume outbound traffic from removable-media executables;
- remove plaintext credentials from shared documents;
- preserve Windows Search indexes during triage because they can retain deleted or absent content.

Validation

9/9 tasks validated.

CASE 05

05

POISONED BRANCH

Software Supply Chain · Linux Implant · Loot
Recovery

GIT / METTLE / XOR / ZIPCRYPTO



15 / 15

VALIDATED

CASE 05

POISONED BRANCH

Software Supply-Chain Compromise and Loot Recovery

Case Context

Poisoned Branch targets Tom Ainsworth through a malicious repository:

```
diogenes-ticket-parser
```

The author metadata identifies:

```
Sebastain Moran  
cbass.Moran@blackpearl2026.htb
```

Runtime payload reconstruction

The malicious logic lived inside:

```
src/ticket_parser/telemetry.py
```

Rather than shipping a recognizable executable, the repository reconstructed its implant at runtime through XOR between:

```
calibration.bin  
diogenes.jpg
```

The output was:

```
/home/Tom/.cache/.ticket-parser/.integrity
```

The file was a Metasploit Mettle payload configured for:

```
tcp://blackpearl2026.htb:31337
```

The same binary matched `reverse.bin` found on the operator infrastructure, providing direct cross-validation of the payload lineage.

Operator activity

Meterpreter history and endpoint audit artifacts reconstruct document discovery, collection and deletion.

The operator searched:

```
search -d ONBOARDING -f *.pdf
```

collected:

```
Gov_HR_Continuity_Emergency_Callout_Roster.pdf
```

and deleted it with:

```
rm Gov_HR_Continuity_Emergency_Callout_Roster.pdf
```

Moran File Exchange

The operator maintained a Flask/Werkzeug node:

```
BlackPearl2026.htb:9999
```

protected by the static cookie:

```
X-Operator-Auth=napoleon_moran_1894
```

The download route allowed path traversal.

`LOOT.zip`

Stolen material was staged in:

```
LOOT.zip  
SHA256 9faf07cbff1f6ed3b01cc6dbfae2d4a330a2460d88c82a0fa7682ff193479b71
```

The archive used ZipCrypto.

An external plaintext copy of the same [README.txt](#) provided a known-plaintext condition. Reproducing the corresponding deflate stream allowed recovery of the internal ZipCrypto keys:

```
85b6bbc1  
27824945  
ce665bee
```

The archive was therefore decrypted without recovering the original password.

Recovered material included the HR roster and the identity/address of **Sarah Kemp**, whose position was the only one marked [REDACTED](#).

Detection Opportunities

- scan internal and third-party repositories for hidden execution inside telemetry/update code;
- alert on Python writing executable content to hidden cache directories followed by `chmod +x`;
- monitor unusual non-standard outbound ports such as 31337 where policy permits;
- correlate document download with immediate deletion;
- detect newly installed archive tooling or sudden staging of encrypted archives on transfer nodes;
- treat weak archive encryption as insufficient protection for sensitive material.

Validation

15/15 tasks validated.

CASE 06

06

SILENT PASSENGER

Android Head Unit · Staged Malware · Mobile Proxy

ANDROID / MQTT / RSA / PROXY



16 / 20

VALIDATED

CASE 06

SILENT PASSENGER

Android Head-Unit Malware and Mobile Proxy Infrastructure

Case Context

Silent Passenger examines an Android 8.1 Allwinner T3/TOPWAY automotive head unit.

The compromise begins with a privileged system application:

```
/system/priv-app/TWCore/TWCore.apk  
SHA256 d7569563cd0491e76d28a1c6a9929234ffffcf035194d2288d888dced7aa11c2
```

TWCore runs with `android.uid.system`, has package-install capability and contains MQTT logic.

MQTT delivery

The affected region uses:

```
tcp://mqtt.car.cardoor.cn:1883  
dofun:dofun666666  
dofun/car/config/#
```

A retained update message delivers:

```
com.tw.jar1  
version 12  
MD5 6c2e34b30da42085240ede53ab6107d4
```

The APK's recovered MD5 exactly matches the integrity value in the MQTT instruction.

Staged reconstruction

`com.tw.jar1` reconstructs its next stage from 30 XOR-obfuscated byte arrays.

The reflective entry point is:

```
com.c.j.qbh.wa  
channel 2039
```

The reconstructed `stage2.jar` has:

```
SHA256 cf5c8c624967775230573a5a552e2e4e2b3653f2362e8c9b66a801e3b251f37c
```

The loader identifies as version `1.7` and contacts:

```
/api/rsaUpdate
```

through RSA-protected traffic.

The response points to:

```
/vr34der34/dex3.68.png
```

which is not an image. After XOR recovery it becomes the control-stage JAR:

```
SHA256 79e01a591c81554b57e0baaf877ca0a1a1f86f39d38973faa1580089b675838d
```

Control and final proxy module

The first validated configuration request-target is:

```
/api/init?configVersion=3.8&rsa=1&channelId=2039
```

The control stage performs device registration, task retrieval and module loading.

The final module is:

```
sh66.io  
SHA256 906734ebb9a274c5c83a22a4475e27354857d7b5b62a4ab6bb5d8d365692e963
```

loaded through:

```
loadlib2:url
```

Its entry point is:

```
com.miyu.transfer.Client.start
```

with method descriptor:

```
(Landroid/content/Context;Ljava/lang/String;Ljava/lang/String;III)V
```

The accepted integer sequence is:

```
9919, 7717, 8818, 20000
```

Mobile proxy role

The final module implements a custom framed relay protocol and accepted sessions against live challenge relay infrastructure.

The decisive conclusion is that the infected vehicle was not simply a beaconing endpoint. It had been converted into a **mobile proxy asset**.

Cross-case Significance

The narrative places the vehicle at **Pavilion Road car park**, which becomes the launch/home location in Iron Feather.

Detection Opportunities

- audit privileged OEM applications with install-package capability;
- monitor hardcoded or unauthenticated MQTT update channels;
- verify cryptographic signing of head-unit updates independently of MD5 values;
- flag APKs that dynamically reconstruct and load DEX/JAR content;
- inspect automotive egress for unapproved MQTT, HTTP and custom relay connections;
- detect repeated module retrieval from disguised image or `.io` paths;
- segment infotainment/head-unit networks from trusted vehicle or enterprise networks where feasible.

Validation

16/20 tasks validated.

Open:

- T13 — canonical runtime UID;
- T14 — canonical task tuple;
- T19 — concrete UID-dependent authentication frame;
- T20 — exact validator coordinates for the parked relay.

CASE 07

07

IRON FEATHER

PX4 UAV Forensics · Custom Cryptography · Flight
Reconstruction

PX4 / AES-GCM / ULOG / MAVLINK

- 01
- 02
- 03
- 04
- 05
- 06
- 07
- 08
- 09

17 / 17

VALIDATED

CASE 07

IRON FEATHER

PX4 UAV Forensics, Custom Cryptography and Flight Reconstruction

Case Context

Iron Feather examines a PX4 SITL quadcopter whose mission datastore and ULog flight record were protected by custom authenticated encryption.

The supplied firmware was not just executable context; it was the key forensic artifact required to recover the evidence.

Encrypted containers

The mission datastore used:

```
PX4DMENC
```

and the flight log:

```
PX4ULENC
```

Both used a 60-byte header with salt, nonce and GCM authentication tag.

Firmware call-sites identified:

```
AES-256-GCM
```

with the first 44 bytes of the container header used as AAD.

Custom KDF

The key-derivation function was located at:

```
RVA 0x170af0
```

It performed:

```
384 rounds
```

of custom 32-bit mixing before building a 104-byte PBKDF2 password.

The standard KDF stage was:

```
PBKDF2-HMAC-SHA256  
8192 iterations  
32-byte output
```

The resulting AES key was:

```
a40ba87b8a0e21d4ead98b917c4bf0f60cc65b25c614b93f107e5ed1e483d6ce
```

The same key successfully validated the AES-GCM tags and decrypted **both** evidence containers. This is the strongest possible cryptographic confirmation that the recovered key, container format and AAD were correct.

Mission intent

The decrypted dataman contained:

```
Active mission bank: 0  
Mission items:      24
```

Item 10 executed:

```
MAV_CMD_DO_SET_ACTUATOR
param1 = 1.0
```

triggering the payload release.

Planned landing:

```
51.49970, -0.16080
```

Actual flight

The accepted event positions came from raw `vehicle_global_position` samples at the relevant timestamps:

```
Takeoff  51.4996985, -0.1607997
Release  51.5035602, -0.1608417
Impact   51.5016938, -0.1620929
```

The release occurred near Hyde Park / Rotten Row.

During the return phase, the aircraft experienced a severe battery condition. At 520.764 seconds from boot, the log recorded:

```
MAV_CMD_INJECT_FAILURE
command 420
motor off
```

followed by:

```
Stopping motors (4095)
```

The drone impacted in Knightsbridge and disarmed at:

```
576.968 s
```

Detection / Safety Opportunities

Iron Feather is better treated as embedded/UAV forensics than forced into enterprise ATT&CK categories.

Useful controls include:

- integrity protection and signed firmware for flight-controller modifications;
- authenticated logging with independently secured keys;
- alerting on `MAV_CMD_INJECT_FAILURE` or equivalent failure-injection commands outside test environments;
- retention of raw telemetry streams, not only smoothed ground-truth output;
- separation of mission intent from flight-state telemetry during post-incident reconstruction.

Validation

17/17 tasks validated.

CASE 08

08

BORROWED NAME

Active Directory Intrusion · AdaptixC2 · Lateral Movement

AD / ADAPTIXC2 / NTLM / SMB



12 / 12

VALIDATED

CASE 08

BORROWED NAME

Active Directory Intrusion and Lateral Movement

Case Context

Borrowed Name begins on [WS02](#) and ends with SYSTEM-level execution on Domain Controller [DC02](#).

The initial removable-media chain was:

```
Salary_Review_Q3_2026.pdf.lnk
↓
docviewer.exe
↓
version.dll
↓
winupdate.exe
↓
AdaptixC2
```

[docviewer.exe](#) side-loaded the malicious [version.dll](#), which launched the AdaptixC2 agent [winupdate.exe](#).

AdaptixC2

The agent communicated with:

```
192.168.56.1:8818
```

using HTTP and header:

```
X-Beacon-Id
```

The workstation beacon identified user:

```
DIOCORE\afenwick
```

Active Directory discovery

The operator performed account, host, LDAP, DNS, domain and trust discovery.

A critical security-descriptor review of [afenwick](#) exposed a [WriteProperty](#) permission on [userPrincipalName](#).

Internal Monologue and ResetNightmare

At 20:44:11 UTC, the attacker executed an Internal Monologue-style credential-access operation and recovered NetNTLMv2 material associated with [afenwick](#).

The operation then abused:

```
CVE-2026-27912
ResetNightmare
```

to temporarily change [afenwick](#)'s UPN toward [jreed](#), reset the Tier-0 account password and restore the original identity.

Recovered credentials included:

```
afenwick:*Seash5lls*
jreed:Aigohng8vai0seish4zi
```

A NewCredentials session with Logon Type 9 was then created.

Lateral movement to DC02

At 20:49:36 UTC, the attacker transferred:

```
\\192.168.56.11\ADMIN$\svc_bkup
```

and temporarily changed the `defragsvc` service binary path to execute the uploaded agent.

The service configuration was restored afterward, reducing residual configuration changes while preserving SYSTEM-level code execution.

At 20:49:57, the second AdaptixC2 beacon checked in from DC02.

Cross-case Significance

The `svc_bkup` binary is the direct artifact bridge into Last Light.

Its hashes include:

```
MD5      ae74305b36c450b8850e34599426d28a
SHA256   1e44c63950dd4fe1ffcc955c08d509132bcaddf87bbb08f5ea622d0533625274
```

Detection Opportunities

- detect suspicious `.lnk` → `signed/decoy EXE` → `side-loaded DLL` chains from removable media;
- alert on Adaptix-like custom headers and unusual POST paths where known;
- monitor rapid `userPrincipalName` modifications followed by privileged password changes;
- correlate NewCredentials Logon Type 9 with administrative SMB activity;
- alert on writes to `ADMIN$` followed by service `ImagePath` modification/start/restore;
- monitor ACLs granting unexpected `WriteProperty` on identity-critical attributes.

Validation

12/12 tasks validated.

CASE 09

09

LAST LIGHT

Memory Forensics · Kerberos Abuse · Resource-Based
Constrained Delegation

MEMORY / KERBEROS / RBCD / LSASS

- 01
- 02
- 03
- 04
- 05
- 06
- 07
- 08
- 09

8 / 10

VALIDATED

CASE 09

LAST LIGHT

Domain Controller Memory Forensics, Kerberos Abuse and RBCD

Case Context

Last Light begins with the post-lateral-movement state of `DC02`.

The evidence set includes a full memory image, Security and Sysmon logs, `ntds.dit`, and system/LSA registry hives.

Reconstructing `svc\_bkup`

Memory analysis identified PID 3376:

```
C:\Windows\svc_bkup
```

The binary was reconstructed from the file-backed memory section:

```
Size      103,936 bytes
MD5       ae74305b36c450b8850e34599426d28a
SHA256    1e44c63950dd4fe1ffcc955c08d509132bcaddf87bbb08f5ea622d0533625274
```

The MD5 exactly matches the artifact deployed in Borrowed Name.

This is one of the strongest cross-case validations in the entire investigation because two independent evidence sources—network/endpoint reconstruction and later memory acquisition—identify the same tool.

Token impersonation

Inside `svc_bkup`, thread 4968 carried an impersonation token associated with the root-domain `rfairfax` RDP session.

The accepted token representation was:

```
0xca8ddc130830
```

with AuthenticationId:

```
0xa616a
```

Kerberos abuse

Memory contained a long-lived root-domain TGT for:

```
Administrator
krbtgt/DIOGENES.HTB
```

with characteristics consistent with a forged Golden Ticket.

The evidence also contained an anomalous service ticket:

```
cifs/DC02.core.diogenes.htb
Administrator@core.diogenes.htb
```

whose PAC included:

```
Extra SID: S-1-18-2
```

consistent with service-asserted identity in an S4U flow.

RBCD

Security log events reconstruct a clear Resource-Based Constrained Delegation sequence.

The attacker created:

```
svc_bkup$  
S-1-5-21-2253468260-689643353-167204612-1140
```

Then `jreed`, under a network logon from `192.168.56.1`, wrote:

```
msDS-AllowedToActOnBehalfOfOtherIdentity
```

on `DC02`.

The subsequent service ticket for Administrator provides memory-level corroboration of the delegation flow.

Detection Opportunities

- alert on Event 5136 modifications of `msDS-AllowedToActOnBehalfOfOtherIdentity`;
- correlate new machine-account creation with delegation changes on Domain Controllers;
- monitor machine-account creation outside approved provisioning workflows;
- investigate long-lived or anomalous Kerberos TGTs in memory;
- correlate service execution with new files in `C:\Windows` or `ADMIN$`;
- inspect EDR/memory telemetry for thread-level privileged token impersonation;
- correlate Logon Type 3/9 activity with directory changes and SMB execution.

Validation

8/10 tasks validated.

Open:

- T2 — expected address/ccache representation for the extra-SID ticket;
- T3 — expected validator representation of the recovered PAC checksum material.

The broader attack reconstruction is not dependent on those two submission formats.

13. Cross-Scenario Correlation

13.1 Silvertown as an operational anchor

Silent Dividend produces a direct geographic value:

```
51.5049, 0.0348
```

Whisper Chain independently places Watson's detention in Silvertown.

Bottle Out examines the laptop of the operator tied to that detention.

The geographic evidence does not prove that every Silvertown reference describes the same physical event, but together it establishes Silvertown as a repeated operational location in the campaign.

13.2 Murknet, `spur` and the Watson custody chain

Whisper Chain identifies:

- Operation Snatch;
- Watson's abduction;
- `dynamite` as the transport actor;
- `spur` as the custody affiliate;
- container `75JM77`;
- immediate cleanup instructions after the site was discovered.

Bottle Out then recovers a real endpoint configured with:

```
spur  
spurio9@murknet.htb  
murknet.htb
```

and identifies the associated operator as Abel Stokes.

This is a cross-source progression from communications intelligence into seized-endpoint forensics.

13.3 Tom Ainsworth: personnel exposure to supply-chain compromise

Paper Ghost recovers Tom Ainsworth's DIOGENES personnel/access record from residual Windows Search data.

Poisoned Branch then shows a ticket-parser repository tailored to the type of work Ainsworth performs.

The later Linux evidence establishes compromise through `diogenes-ticket-parser`, runtime reconstruction of the Mettle payload and collection of sensitive HR material.

This is a particularly useful case-study chain because it shows how an information exposure can become operationally relevant later even when the initial compromise mechanism differs.

13.4 Mobile relay to aerial mission

Silent Passenger concludes that the compromised vehicle itself functions as a mobile proxy relay and places the vehicle at Pavilion Road.

Iron Feather's accepted takeoff/home coordinate is:

```
51.4996985, -0.1607997
```

corresponding to the same Pavilion Road location.

The two cases therefore form a coherent mobile-infrastructure thread: compromised vehicle relay followed by an aerial mission launched from the same area.

13.5 `svc\_bkup`: artifact-level continuity

Borrowed Name records the upload of:

```
\\192.168.56.11\ADMIN$\svc_bkup
```

with:

```
MD5 ae74305b36c450b8850e34599426d28a
```

Last Light reconstructs `C:\Windows\svc_bkup` directly from DC02 memory and recovers the same MD5.

This is not merely narrative continuity. It is direct file identity across independent evidence sets.

14. Unified MITRE ATT&CK View

The table below consolidates only techniques directly supported by the case evidence. Iron Feather is intentionally not forced into the Enterprise ATT&CK model because its most important behaviors are mission-control and UAV-forensic events.

Technique	ID	Cases	Evidence
Replication Through Removable Media	T1091	S04, S08	Planted/malicious USB delivery
Supply Chain Compromise: Software Supply Chain	T1195.002	S05	Poisoned ticket-parser repository
User Execution: Malicious File	T1204.002	S01, S04, S05, S08	User-triggered installer/update/lure execution
PowerShell	T1059.001	S01, S02	Hidden implant launch; Operation Vanish
Python	T1059.006	S05	Malicious <code>telemetry.py</code> execution path
DLL Side-Loading	T1574.002	S08	<code>docviewer.exe</code> → malicious <code>version.dll</code>
Obfuscated Files or Information	T1027	S01, S05, S06	Lua strings, XOR payloads, disguised stages
Deobfuscate/Decode Files or Information	T1140	S01, S06	Runtime decryption/reconstruction
Masquerading	T1036 / T1036.005	S04, S05	Fake update package; disguised Linux payload components
File Deletion	T1070.004	S02, S03, S05	Operation Vanish and source-document deletion
Clear Windows Event Logs	T1070.001	S02	Security log clear event
Credentials In Files	T1552.001	S01, S04	Wallet secrets and DIOGENES credentials
Audio Capture	T1123	S04	Microphone access
Video Capture	T1125	S04	Webcam access
Data from Local System	T1005	S04, S05	Sensitive local documents
Archive Collected Data	T1560.001	S05	<code>L00T.zip</code>
Account Discovery	T1087	S08	LDAP user enumeration
Remote System Discovery	T1018	S08	Domain-host discovery
System Network Configuration Discovery	T1016	S08	<code>ipconfig</code> tasking
Application Layer Protocol / Web Protocols	T1071 / T1071.001	S01, S02, S03, S06, S08	WinHTTP, HTTPS, XMPP, MQTT/HTTP, Adaptix
Encrypted Channel	T1573 / T1573.001	S03, S06	AES-protected XMPP dispatcher; RSA tasking
Ingress Tool Transfer	T1105	S06	APK/module retrieval

Technique	ID	Cases	Evidence
Proxy	T1090	S06	Vehicle used as mobile relay
Non-Standard Port	T1571	S05, S06	Mettle and relay infrastructure
Remote Access Software	T1219	S02	Tactical RMM
External Remote Services	T1133	S02	OpenVPN access
Windows Service	T1543.003	S08, S09	defragsvc abuse and service-based execution
SMB/Windows Admin Shares	T1021.002	S08, S09	ADMIN\$ transfer
Account Manipulation	T1098	S08, S09	UPN/password manipulation and RBCD-related directory changes
Token Impersonation/Theft	T1134.001	S09	rfaifax token inside svc_bkup
Golden Ticket	T1558.001	S09	Forged long-lived root-domain TGT
Domain Account	T1136.002	S09	svc_bkup\$ machine-account creation

15. Indicators and Detection Engineering

15.1 High-value infrastructure

Murknet / XMPP

```
murknet.htb
groups.murknet.htb
command.murknet.htb
upload.murknet.htb
```

Bottle Out

```
api.antimattercommunication.xyz
18.156.81.166:7577
NPLN-CA
spurio9@murknet.htb
```

Poisoned Branch

```
blackpearl2026.htb:31337
BlackPearl2026.htb:9999
X-Operator-Auth=napoleon_moran_1894
```

Silent Passenger

```
mqt.ccar.cardoor.cn:1883
a1.ishano456.sbs
144.217.243.201
199.190.45.202:9919
23.251.102.195:9999
```

Borrowed Name / Last Light

```
192.168.56.1:8818
WS02 192.168.56.22
DC02 192.168.56.11
```

15.2 High-value file hashes

```
Silent Dividend – TrustSettle 1.0.0.exe
SHA256 c366e00a4ac1b4df56d1e4e7bb94e1c10937f86cffde733424fb7c7dd5a444fc

Poisoned Branch – .integrity
SHA256 f925cafffe2310ae74ac936b78c0fd5d64b249a5be57e73c28ee6b262e5d44c79

Silent Passenger – stage2.jar
SHA256 cf5c8c624967775230573a5a552e2e4e2b3653f2362e8c9b66a801e3b251f37c

Silent Passenger – control stage
SHA256 79e01a591c81554b57e0baaf877ca0a1a1f86f39d38973faa1580089b675838d

Silent Passenger – sh66.io
SHA256 906734ebb9a274c5c83a22a4475e27354857d7b5b62a4ab6bb5d8d365692e963

Borrowed Name / Last Light – svc_bkup
SHA256 1e44c63950dd4fe1ffcc955c08d509132bcaddf87bbb08f5ea622d0533625274
MD5 ae74305b36c450b8850e34599426d28a
```

15.3 Detection strategy by layer

Endpoint execution

High-signal correlations include:

```
powershell.exe
→ luajit.exe
→ api.txt
```

and:

```
removable media
→ .lnk
→ decoy executable
→ side-loaded DLL
→ C2 agent
```

These chains are more useful than detecting any one executable name in isolation.

Identity and Active Directory

Priority detections:

- `userPrincipalName` modification followed by privileged password reset;
- new machine account creation outside provisioning workflows;
- Event 5136 on `msDS-AllowedToActOnBehalfOfOtherIdentity`;
- NewCredentials Logon Type 9 near SMB/admin-share activity;
- service ImagePath modification followed by service start and rapid restoration;
- privileged token impersonation inside non-standard service processes.

Messaging and command infrastructure

Where XMPP is not a business requirement:

- outbound XMPP/STARTTLS should be rare and highly reviewable;
- service discovery of MUC/pubsub/upload components should be monitored;
- dedicated external XMPP clients on operational endpoints are high-value findings.

For MQTT-enabled embedded platforms:

- privileged update subscriptions should require cryptographic authorization;
- update payloads should be signed, not merely MD5-checked;

- brokers should not expose fleet-wide retained instructions through static shared credentials.

Data collection and exfiltration

Correlate:

- sensor access by unusual desktop executables;
- document collection followed by immediate deletion;
- creation of encrypted staging archives;
- outbound volume anomalies from executables launched from removable media;
- use of transfer nodes with weak static-cookie authentication.

Kerberos and delegation

Review:

- anomalously long TGT lifetimes;
- unexpected RC4 in otherwise modern environments;
- S4U tickets associated with recently modified delegation attributes;
- S-1-18-2 in context with RBCD changes;
- machine-account creation and service-ticket acquisition in short succession.

16. Analytical Lessons

16.1 Secondary artifacts can be primary evidence

Paper Ghost is the clearest example. The original DIOGENES documents were absent, but Windows Search preserved enough content to reconstruct them.

Similarly, Bottle Out demonstrates that deleting application directories does not erase NTFS metadata, browser storage, registry state or event traces.

16.2 Cross-validation is more valuable than single-tool certainty

The strongest conclusions were rarely based on one parser.

Examples:

- Iron Feather's KDF was validated by both reverse engineering and successful AES-GCM authentication;
- Poisoned Branch linked victim and operator payloads by identical binary identity;
- Borrowed Name and Last Light independently observed the same `svc_bkup` artifact;
- Paper Ghost correlated execution across UserAssist, ConsentStore and SRUM.

16.3 Platform acceptance and forensic truth are related but not identical

Silent Passenger reconstructed several runtime behaviors correctly even where canonical challenge values remained unresolved.

Last Light identified the relevant Kerberos ticket and PAC checksum material even though Tasks 2 and 3 remained open because the expected submission representation was unknown.

Whisper Chain similarly established the latest operation and objective while Task 8 remained unresolved at the validator level.

A mature investigation preserves that distinction rather than converting every technically supported finding into a claimed solved flag.

16.4 Dynamic infrastructure requires state-aware analysis

Silent Passenger demonstrated that registration IDs, host families, task sets and order identifiers changed across live runs.

This is a useful real-world lesson: when an infrastructure component is stateful or shared, a later replay may not reproduce the exact value observed during the original event.

16.5 Mission intent and observed behavior must be separated

Iron Feather contained both a planned mission and a real flight.

The mission datastore describes intent. The ULog describes execution.

That same principle applies more broadly to endpoint and directory forensics: configuration alone is not proof of execution, and a recovered tool is not proof that every capability was used.

16.6 Anti-forensics often create their own timeline

Operation Vanish and the deletion of the HR roster are examples where cleanup activity becomes a high-value event.

Deletion itself establishes operator intent, narrows the incident window and tells the investigator which artifacts the attacker considered sensitive.

17. Conclusion

The Reichenbach Directive is best understood as a campaign-scale investigation assembled from heterogeneous evidence.

The nine Sherlocks required different technical methods, but the same analytical principles held throughout:

1. preserve the evidence;
2. identify the artifact that can actually answer the question;
3. distinguish direct fact from correlation and inference;
4. validate high-impact conclusions through an independent source where possible;
5. retain uncertainty when the evidence does not support a stronger statement.

The result is a connected case study spanning:

```
Windows endpoint compromise
→ XMPP coordination
→ physical intrusion
→ software supply chain
→ Linux C2 and exfiltration
→ automotive Android malware
→ mobile proxy infrastructure
→ UAV mission forensics
→ Active Directory lateral movement
→ Kerberos and RBCD abuse
```

The final competition result was:

Exploit Bag Chaser (solo participant/team of one) · #211 / 5,637 teams · 104 / 111 flags · 8,200 points

with six Sherlocks completely solved.

The strongest portfolio value is not the flag count alone. It is the breadth of evidence types and the ability to correlate them without sacrificing forensic precision.

Appendix A — Complete 111-Task Validation Matrix

The matrix below preserves the final status of every task across the nine Sherlocks.

Case	Task	Investigative Question / Result	Answer / Finding	Status
S01 — Silent Dividend	1	—	C:\Users\Public	Validated
S01 — Silent Dividend	2	—	FILE_NOTIFY_INFORMATION	Validated
S01 — Silent Dividend	3	—	WinHttpRequest	Validated
S01 — Silent Dividend	4	—	resolveState()	Validated
S01 — Silent Dividend	5	—	AUTH=NAPOLEON SETTLEMENT_REFERENCE=SR-4821	Validated
S01 — Silent Dividend	6	—	TEMP	Validated
S01 — Silent Dividend	7	—	approve()	Validated
S01 — Silent Dividend	8	—	115792089237316195423570985008687907 853269984665640564039457584007913129 639935	Validated
S01 — Silent Dividend	9	—	BrowserProvider	Validated
S01 — Silent Dividend	10	—	51.5049,0.0348	Validated
S02 — Bottle Out	1	VPN remote address and port	18.156.81.166:7577	Validated
S02 — Bottle Out	2	VPN client certificate CA	NPLN-CA	Validated
S02 — Bottle Out	3	VPN-assigned IP	10.129.175.2	Validated
S02 — Bottle Out	4	Remote-management agent and version	Tactical RMM Agent v.2.11.0	Validated
S02 — Bottle Out	5	Agent C2 domain	api.anti-mattercommunication.xyz	Validated
S02 — Bottle Out	6	Agent authentication token	98ec588da683c01820232943a6151e8e7772 419b	Validated
S02 — Bottle Out	7	First Operation Vanish command	Remove-Item -LiteralPath C:\Users\ \spur\Gajim -Recurse -Force	Validated
S02 — Bottle Out	8	IM account	spurio9@murknet.htb	Validated
S02 — Bottle Out	9	IM password	spur999!*	Validated
S02 — Bottle Out	10	Full name of the jailer	Abel Stokes	Validated
S03 — Whisper Chain	1	—	murknet.htb,command.murknet.htb,grou ps.murknet.htb,upload.murknet.htb	Validated
S03 — Whisper Chain	2	—	Infrastructure,Random,Resources,Rule s	Validated
S03 — Whisper Chain	3	—	zytglogge88@murknet.htb:TickTock24!	Validated
S03 — Whisper Chain	4	—	https://stonedforums.htb/@porlock	Validated

Case	Task	Investigative Question / Result	Answer / Finding	Status
S03 — Whisper Chain	5	—	429x3WVq1ucARGXx6NEwL4Sg4iowfW5ZWMAq EDerLxrwdg4ffkonB5tNyg85BK6jDqDQRfBE RANhgf6DnGjjFyDR5L7uwy	Validated
S03 — Whisper Chain	6	—	dynamite	Validated
S03 — Whisper Chain	7	—	Victoria Station	Validated
S03 — Whisper Chain	8	—	Evidence supports Operation Snatch / kidnapping Watson; validator representation unresolved	Open
S04 — Paper Ghost	1	—	2026-08-19 15:35:50	Validated
S04 — Paper Ghost	2	—	RS200000000627E4&0	Validated
S04 — Paper Ghost	3	—	E:\C0-LT-0469 update package\ \update.exe	Validated
S04 — Paper Ghost	4	—	2026-08-19 15:36:25	Validated
S04 — Paper Ghost	5	—	C0-USB-0091	Validated
S04 — Paper Ghost	6	—	2026-08-19 15:38:08	Validated
S04 — Paper Ghost	7	—	127	Validated
S04 — Paper Ghost	8	—	172.064531	Validated
S04 — Paper Ghost	9	—	tainsworth:D10g3n3s_T1ck3ts#2026	Validated
S05 — Poisoned Branch	1	—	diogenes-ticket-parser	Validated
S05 — Poisoned Branch	2	—	cbass.Moran@blackpearl2026.htb	Validated
S05 — Poisoned Branch	3	—	calibration.bin	Validated
S05 — Poisoned Branch	4	—	/home/Tom/.cache/.ticket- parser/.integrity	Validated
S05 — Poisoned Branch	5	—	31337	Validated
S05 — Poisoned Branch	6	—	1514	Validated
S05 — Poisoned Branch	7	—	BlackPearl2026.htb:9999	Validated
S05 — Poisoned Branch	8	—	X-Operator-Auth=napoleon_moran_1894	Validated
S05 — Poisoned Branch	9	—	rm Gov_HR_Continuity_Emergency_Callout_ Roster.pdf	Validated
S05 — Poisoned Branch	10	—	1549	Validated
S05 — Poisoned Branch	11	—	set payload linux/x64/meterpreter_reverse_tcp	Validated
S05 — Poisoned Branch	12	—	search -d ONBOARDING -f *.pdf	Validated
S05 — Poisoned Branch	13	—	L00T.zip	Validated
S05 — Poisoned Branch	14	—	Sarah Kemp	Validated

Case	Task	Investigative Question / Result	Answer / Finding	Status
S05 — Poisoned Branch	T5	—	Flat 6, Ashdown House, Palace Court, London W2 4LS	Validated
S06 — Silent Passenger	T1	—	/system/priv-app/TwCore/ TwCore.apk:d7569563cd0491e76d28a1c6a9929234ffffcf035194d2288d888dced7aa11c2	Validated
S06 — Silent Passenger	T2	—	ro.com.google.gmsversion	Validated
S06 — Silent Passenger	T3	—	tcp://mqtt.car.cardoor.cn:1883 dofun:dofun666666	Validated
S06 — Silent Passenger	T4	—	dofun/car/config/#	Validated
S06 — Silent Passenger	T5	—	com.tw.jar1:12:6c2e34b30da42085240ed e53ab6107d4	Validated
S06 — Silent Passenger	T6	—	com.c.j.qbh.wa:2039	Validated
S06 — Silent Passenger	T7	—	cf5c8c624967775230573a5a552e2e4e2b36 53f2362e8c9b66a801e3b251f37c	Validated
S06 — Silent Passenger	T8	—	1.7	Validated
S06 — Silent Passenger	T9	—	/api/rsaUpdate	Validated
S06 — Silent Passenger	T10	—	/vr34der34/dex3.68.png	Validated
S06 — Silent Passenger	T11	—	79e01a591c81554b57e0baaf877ca0a1a1f8 6f39d38973faa1580089b675838d	Validated
S06 — Silent Passenger	T12	—	/api/init? configVersion=3.8&rsa=1&channelId=2039	Validated
S06 — Silent Passenger	T13	—	Runtime UID mechanism reconstructed; canonical value unresolved	Open
S06 — Silent Passenger	T14	—	Task structure reconstructed; canonical tuple unresolved	Open
S06 — Silent Passenger	T15	—	loadlib2:url	Validated
S06 — Silent Passenger	T16	—	906734ebb9a274c5c83a22a4475e27354857 d7b5b62a4ab6bb5d8d365692e963	Validated
S06 — Silent Passenger	T17	—	(Landroid/content/Context;Ljava/ lang/String;Ljava/lang/String;III)V	Validated
S06 — Silent Passenger	T18	—	9919, 7717, 8818, 20000	Validated
S06 — Silent Passenger	T19	—	Authentication framing reconstructed; canonical UID-dependent frame unresolved	Open
S06 — Silent Passenger	T20	—	Pavilion Road car park identified; exact validator coordinates unresolved	Open
S07 — Iron Feather	1	—	PX4DMENC	Validated
S07 — Iron Feather	2	—	AES-256-GCM	Validated
S07 — Iron Feather	3	—	0x170af0	Validated

Case	Task	Investigative Question / Result	Answer / Finding	Status
S07 — Iron Feather	4	—	384	Validated
S07 — Iron Feather	5	—	PBKDF2-HMAC-SHA256	Validated
S07 — Iron Feather	6	—	a40ba87b8a0e21d4ead98b917c4bf0f60cc65b25c614b93f107e5ed1e483d6ce	Validated
S07 — Iron Feather	7	—	0	Validated
S07 — Iron Feather	8	—	24	Validated
S07 — Iron Feather	9	—	10	Validated
S07 — Iron Feather	10	—	51.49970, -0.16080	Validated
S07 — Iron Feather	11	—	51.4996985, -0.1607997	Validated
S07 — Iron Feather	12	—	51.5035602, -0.1608417	Validated
S07 — Iron Feather	13	—	MAV_CMD_INJECT_FAILURE	Validated
S07 — Iron Feather	14	—	277	Validated
S07 — Iron Feather	15	—	51.5016938, -0.1620929	Validated
S07 — Iron Feather	16	—	576.968	Validated
S07 — Iron Feather	17	—	Knightsbridge	Validated
S08 — Borrowed Name	1	—	Adaptix	Validated
S08 — Borrowed Name	2	—	53fc4c03c7b461befe5dcb268e3d9208:4580221ac3fe51be1797524a048e552d	Validated
S08 — Borrowed Name	3	—	CVE-2026-27912	Validated
S08 — Borrowed Name	4	—	583236cc3ef2488fb133385bcd75825e	Validated
S08 — Borrowed Name	5	—	S-1-5-21-2253468260-689643353-167204612-1125	Validated
S08 — Borrowed Name	6	—	internal_monologue:2026-09-09 20:44:11	Validated
S08 — Borrowed Name	7	—	afenwick:*Seash5lls*	Validated
S08 — Borrowed Name	8	—	jreed:Aigohng8vai0seish4zi	Validated
S08 — Borrowed Name	9	—	9	Validated
S08 — Borrowed Name	10	—	\\\\192.168.56.11\\ADMIN\$\\svc_bkup	Validated
S08 — Borrowed Name	11	—	defragsvc	Validated
S08 — Borrowed Name	12	—	ddc68fa7:289122cf1ec91c67eb89c30642adfea4	Validated
S09 — Last Light	1	—	1e44c63950dd4fe1ffcc955c08d509132bcaddf87bbb08f5ea622d0533625274	Validated
S09 — Last Light	2	—	Ticket with extra SID identified; expected address/ccache representation unresolved	Open

Case	Task	Investigative Question / Result	Answer / Finding	Status
S09 — Last Light	3	—	PAC checksums extracted; expected validator representation unresolved	Open
S09 — Last Light	4	—	0xca8ddc130830	Validated
S09 — Last Light	5	—	2:0xa616a	Validated
S09 — Last Light	6	—	4968	Validated
S09 — Last Light	7	—	svc_bkup\$:S-1-5-21-2253468260-689643353-167204612-1140	Validated
S09 — Last Light	8	—	S-1-5-21-3066635835-107521988-671693545	Validated
S09 — Last Light	9	—	0x3f213b:3	Validated
S09 — Last Light	10	—	0x17c514:0x17c569	Validated

Appendix B — Unresolved Tasks

Case	Task	Current evidentiary state
Whisper Chain	T8	Evidence supports Operation Snatch / kidnapping Watson ; validator representation unresolved
Silent Passenger	T13	Runtime UID mechanism reconstructed; canonical accepted UID unresolved
Silent Passenger	T14	Task structure reconstructed; canonical accepted tuple unresolved
Silent Passenger	T19	Authentication-frame format reconstructed; canonical UID-dependent frame unresolved
Silent Passenger	T20	Pavilion Road car park identified; exact accepted coordinate pair unresolved
Last Light	T2	Relevant extra-SID ticket identified; expected address/ccache representation unresolved
Last Light	T3	PAC checksum material extracted; expected validator representation unresolved

Appendix C — Audited Source Report Integrity

The master report was generated from the nine audited case reports listed below.

Source report	SHA-256
S01_Silent_Dividend_FINAL_AUDITED.md	866c5d7671dbedbf159bffac3389d0241f4206e25b6454175bc9390a77ec352f
S02_Bottle_Out_FINAL_AUDITED.md	667657954dddc9bfff6f92f2cb6cd6337050d2504209e71ed736cecb3ef5b1608
S03_Whisper_Chain_FINAL_AUDITED.md	687d8dce480c0390d9953a4991810c6cb22070ed096012c8acb3bc18435b9d5b
S04_Paper_Ghost_FINAL_AUDITED.md	7141c0c45d9c388f89a9d88d361548f100ac30c7722cd987117d849a6de2a4cf
S05_Poisoned_Branch_FINAL_AUDITED.md	a5fae86361e95ac21330acac94efae172b1140306a8f81bbe198de77b35a088b
S06_Silent_Passenger_FINAL_AUDITED.md	d8981c05578ce7ade332149001ea77535ffe356b975d7b571374a23bfa420a21
S07_Iron_Feather_FINAL_AUDITED.md	131ca6693a1741c544aa7db77f696e604f33983dd0daac51f80b7d662eeab01f
S08_Borrowed_Name_FINAL_AUDITED.md	0f068af812f0e4fcb115ec59d43c12ea9231f1bd7d81267c2f983ccf504c8581
S09_Last_Light_FINAL_AUDITED.md	fa775b188a84c10bb72f9f9337acee36d823c495938534f01148a7f32abd16ee

Appendix D — Master Report Design Specification

The PDF/layout stage should preserve the following visual system:

Element	Specification
Editorial paper	#F8FBFB
Cover deep teal / blue-grey	#0B2730

Element	Specification
Body text	#0A2730
Secondary text	#6D7B82
Primary teal	#188EA0
Deep teal	#0B5665
Technical blue	#2B7A9D
Validated green	#2F855A
Open / warning amber	#B7791F
Code / evidence panel	#0D3039

Recommended composition:

- dark blue-grey / forensic teal cover (#0B2730) aligned to the final cyber-investigative artwork; pure black is avoided while preserving a Blue Team / DFIR identity;
- light body pages for long-form readability;
- light chapter openers with large low-contrast case numbers;
- integrated cyber-investigative overlays on the cover: node-link analysis, London map traces, packet-flow motifs, endpoint/server/mobile/drone/AD pivots and a restrained evidence progression;
- visual motifs must remain evidence-driven and semantically tied to investigation, infrastructure, telemetry or attack progression; no floating report pages or arbitrary decorative geometry;
- monospace treatment for hashes, SIDs, paths, addresses and protocol values;
- teal as the primary Blue Team / DFIR identity color;
- technical blue for pivots, correlation paths and secondary evidence;
- dark panels reserved for code/evidence blocks; the only full-page dark treatment is the cover;
- green only for validated status;
- amber only for unresolved or cautionary states.

Cover copy:

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HOLMES CTF 2026

THE REICHENBACH DIRECTIVE
A Complete DFIR Investigation of Operation DIOGENES

Michel-DV
HTB · InFracturaVeritas
DFIR · Threat Hunting · Reverse Engineering

EXPLOIT BAG CHASER · #211 / 5,637 TEAMS · 104 / 111 FLAGS · 8,200 POINTS
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The first interior spread should carry:

Nine investigations. One operation.

followed by the incident map and master timeline.

OFFICIAL COMPETITION CERTIFICATE



Hack The Box HOLMES CTF 2026 · 17/09/2026-21/09/2026 · Team Rank 211 / 5,637 · 104/111 · 8,200 points